

The students decide not to use the bridge rectifier in the circuit and return to the original design of Figure 1. They next test the smoothing operation with a modified circuit shown in Figure 5 using $C = 2000\ \mu\text{F}$, a switch, S , and a $100\ \Omega$ resistor. The switch is initially **open** so the resistor is disconnected at this time.

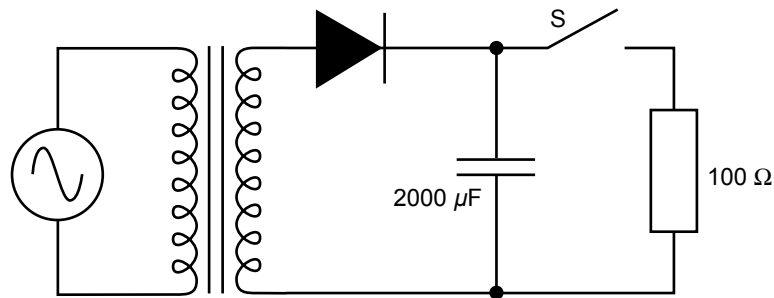


Figure 5

Question 5

The time constant of the circuit shown in Figure 5 is measured by the students to be 2 ms. What is the effective forward resistance of the diode?

- A. $0.1\ \Omega$
- B. $1\ \Omega$
- C. $2\ \Omega$
- D. $10\ \Omega$

2 marks

The students now **close** the switch of the circuit of Figure 5, connecting the $100\ \Omega$ resistor into the circuit. The voltage across the $100\ \Omega$ resistor is measured using a CRO and is observed to comprise a DC component of about 12 V and an AC component, or ripple. They investigate this ripple by varying the values of some of the components in the circuit of Figure 5.

Question 6

Which one of the following statements (A.–D.) concerning the magnitude of the ripple voltage is correct?

- A. It approximately doubles when the capacitance is halved to $1000\ \mu\text{F}$ and the resistance doubled to $200\ \Omega$.
- B. It approximately doubles when the capacitance is doubled to $4000\ \mu\text{F}$ and the resistance halved to $50\ \Omega$.
- C. It approximately doubles when the capacitance is halved to $1000\ \mu\text{F}$ and the resistance left unaltered at $100\ \Omega$.
- D. It approximately doubles when the capacitance is doubled to $4000\ \mu\text{F}$ and the resistance left unaltered at $100\ \Omega$.

2 marks

The students now consider the final part of the design, the voltage regulator, using a resistor, R_V , in series with a 9 V Zener diode (see Figure 1).

The current-voltage characteristic of the Zener diode is shown in Figure 6.

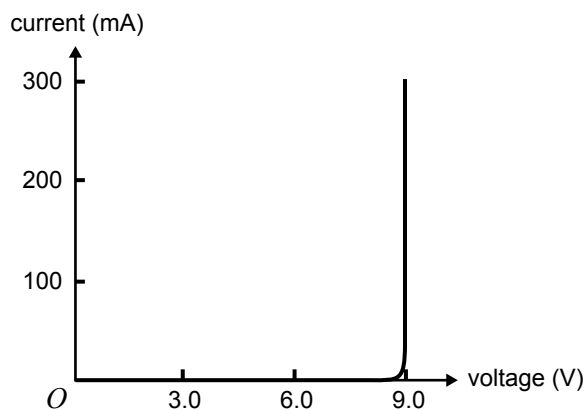


Figure 6

The students use a multimeter to measure the DC voltage across the $30\ \Omega$ resistor in Figure 1 to be approximately 3 V.

Question 7

Given that the voltage across the transformer is 24 V (peak-to-peak), in the space below explain why this value of 3 V across R_V is to be expected.

2 marks

Question 8

In this measurement for Question 7 the students could have used either a battery-powered multimeter (MM) or the mains-powered CRO.

Which one of the following statements concerning these two instruments is true?

- A. The MM (in DC voltage mode) is the most sensible choice to measure this voltage across R_V compared to a CRO as it is less likely to affect the power supply circuit.
- B. The CRO is the better voltage measuring instrument in this situation, as it can measure both the AC and DC voltages across R_V , while a MM can only measure DC voltages.
- C. The MM is the better instrument to use as the only way to measure a current through a resistor is by using a MM (in AC or DC current mode). When using a CRO there is no way that the current in the resistor can be determined.
- D. The CRO is the better instrument as it can be used to fully describe both the AC and DC voltages, while the MM cannot.

2 marks

Question 9

What is the power dissipated in the $30\ \Omega$ resistor, R_V , of Figure 1? Express your answer in mW.

mW

2 marks

The students now have a fully operational regulated 9 V DC power supply as shown in Figure 1. The final circuit has $C = 2000\ \mu\text{F}$, a 9 V Zener diode regulator and $R_V = 30\ \Omega$. To start with, the students connect a load resistor of $R_L = 300\ \Omega$. They then do a series of tests to observe what might change the output DC across the load resistor.

Question 10

Based on their observations, which one of the following statements (**A.–D.**) is correct?

- A.** The output DC voltage changes if the mains supply voltage is increased to 250 V (RMS).
- B.** The output DC voltage changes if the load resistor is increased to $400\ \Omega$.
- C.** The output DC voltage changes if the 9 V Zener diode is replaced by a 6 V Zener diode.
- D.** The output DC voltage changes if the capacitor is increased to $3000\ \mu\text{F}$.

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2 marks

Question 11

The students have their circuit checked by a qualified electrician who advises them that their power supply circuit will require a heat sink. Explain why a heat sink is used in this regulated power supply circuit, and what part of the circuit needs to be attached to this heat sink.

2 marks

As a final part of their project to design and construct a regulated DC power supply, the students have to write a brief report. Students discuss safety and risk associated with this project.

Question 12

Which is the most important activity to ensure maximum safety and minimum risk?

- A. learning from trial and error when designing and constructing electrical circuits
- B. getting the teacher (or someone appropriately qualified) to check each stage of the design and construction
- C. understanding how to correctly use a cathode ray oscilloscope and a multimeter for troubleshooting and diagnosis
- D. understanding the physics of specific components such as the transformer, diodes, capacitor and resistor

2 marks